

CO4 ASSESSMENT TOOL 1 – LEVEL 2

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
C PROGRAMMING

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10/10/2020

CO4 - AT1 - LEVEL 2 Exam

Level Exam · C · 02 Sep 2020, 10:00 AM

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Result Available
Your score: 80 / 100**QUESTIONS**
LEVEL 2 — MEDIUM
Q1
1D array
Q2
structure
Q3
string
Q4
2D array
4 / 4 saved**Level 2 — Medium: triangular matrix (M)** — 2D arrayWrite a program to find the upper triangular matrix of a given matrix.
Sample Input:
Enter number of rows: 3
Enter number of columns: 3
Enter elements:
1 2 3
4 5 6
7 8 9
Sample Output:
Upper triangular matrix:
1 2 3
0 5 6
0 0 9**Your Code**

```
1 #include <stdio.h>
2
3 int main() {
4     int rows, cols, i, j;
5
6     printf("Enter number of rows: ");
7     scanf("%d", &rows);
8     printf("Enter number of column: ");
9     scanf("%d", &cols);
10
11     int matrix[rows][cols];
12
13     printf("Enter elements:\n");
14     for (i = 0; i < rows; i++) {
15         for (j = 0; j < cols; j++) {
16             scanf("%d", &matrix[i][j]);
17         }
18     }
19
20     for (i = 0; i < rows; i++) {
21         for (j = 0; j < cols; j++) {
22             if (i > j) {
23                 matrix[i][j] = 0;
24             }
25         }
26     }
27
28     printf("\nUpper triangular matrix:\n");
29     for (i = 0; i < rows; i++) {
30         for (j = 0; j < cols; j++) {
31             printf("%d ", matrix[i][j]);
32         }
33         printf("\n");
34     }
35
36     return 0;
37 }
```

Program Input (stdin)
Enter Input If required...**Output**
Enter number of rows: Enter number of
columns: Enter elements:
Upper triangular matrix:
1 2 3

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QUESTIONS

LEVEL 2 — MEDIUM

Q1

Theory

Q2

Structure

Q3

Programming

Q4

Theory

4 / 4 saved

Level 2 — Medium student (M) — Structure

Develop a university course management system. Define structures for Course, Professor, Student, and Enrollment. Write a program to allow professors to add courses, students to enroll in courses, and grades to be assigned.

Sample Input/Output:

Welcome to the University Course Management System!

Menu:

1. Add New Course
2. Enroll Student in Course
3. Assign Grade

Enter your choice: 1

Enter course name: Computer Science
Enter professor name: Dr. Johnson
Enter course credits: 3

Course added successfully!

Menu:

1. Add New Course
2. Enroll Student in Course
3. Assign Grade

Enter your choice: 2

Enter student name: Alice
Enter course name to enroll: Computer Science

Student enrolled in course successfully!

Menu:

1. Add New Course
2. Enroll Student in Course
3. Assign Grade

Enter your choice: 3

Enter student name: Alice
Enter course name: Computer Science
Enter grade: A

Grade assigned successfully!

Your Code

```
1 #include <stdio.h>
2 #include <string.h>
3
4 struct Professor {
5     char name[50];
6 };
7
8 struct Course {
9     char courseName[50];
10    struct Professor professor;
11    int credits;
12 };
13
14 struct Student {
15     char name[50];
16 };
17 struct Enrollment {
18     struct Student student;
19     char courseName[50];
20     char grade[5];
21 };
22
23 struct Course course;
24 struct Enrollment enrollment;
25
26 int main() {
27     int choice;
28     printf("Welcome to the University Course Management System!\n");
29
30     while (1) {
31         printf("\nMenu:\n");
32         printf("1. Add New Course\n");
33         printf("2. Enroll Student in Course\n");
34         printf("3. Assign Grade\n");
35         printf("4. Exit\n");
36         printf("Enter your choice: ");
37         scanf("%d", &choice);
38         getchar();
39
40         switch (choice) {
41             case 1:
42                 printf("Enter course name: ");
43                 fgets(course.courseName, sizeof(course.courseName), stdin);
44                 strcpy(course.courseName, course.courseName);
45
46                 printf("Enter professor name: ");
47                 fgets(course.professor.name, sizeof(course.professor.name), stdin);
48                 strcpy(course.professor.name, course.professor.name);
49
50                 printf("Enter course credits: ");
51                 scanf("%d", &course.credits);
52                 getchar();
53
54                 printf("Course added successfully!\n");
55                 break;
56             case 2:
57                 printf("Enter student name: ");
58                 fgets(enrollment.student.name, sizeof(enrollment.student.name), stdin);
59                 strcpy(enrollment.student.name, enrollment.student.name);
60
61                 printf("Enter course name to enroll: ");
62                 fgets(enrollment.courseName, sizeof(enrollment.courseName), stdin);
63                 strcpy(enrollment.courseName, enrollment.courseName);
64
65                 printf("Student enrolled in course successfully!\n");
66                 break;
67             case 3:
68                 printf("Enter student name: ");
69                 fgets(enrollment.student.name, sizeof(enrollment.student.name), stdin);
70                 strcpy(enrollment.student.name, enrollment.student.name);
71
72                 printf("Enter course name: ");
73                 fgets(enrollment.courseName, sizeof(enrollment.courseName), stdin);
74                 strcpy(enrollment.courseName, enrollment.courseName);
75
76                 printf("Enter grade: ");
77                 fgets(enrollment.grade, sizeof(enrollment.grade), stdin);
78                 strcpy(enrollment.grade, enrollment.grade);
79
80                 printf("Grade assigned successfully!\n");
81                 break;
82             case 4:
83                 printf("Exiting...\n");
84                 return 0;
85             default:
86                 printf("Invalid choice!\n");
87         }
88     }
89     return 0;
90 }
```

Program Input (stdin)

Enter input if required...

Output

Security Error: Disallowed code pattern detected.

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Result Available
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QUESTIONS

LEVEL 2 — MEDIUM

Q1
101wayQ2
114structureQ3
114stringQ4
121way

4 / 4 saved

Level 2 — MEDIUM interleaving (M) — string

Write a program to check if a given string is formed by interleaving two other strings.

Sample Input:

Enter first string: ABCD

Enter second string: XY

Enter interleaved string to check: AXBYCD

Sample Output:

The string is formed by interleaving the given strings.

Your Code

```
#include <stdio.h>
#include <string.h>

int main() {
    char a1[200], a2[200], a3[400];
    int n1, n2, n3;

    printf("Enter first string: ");
    scanf("%199s", a1);
    printf("Enter second string: ");
    scanf("%199s", a2);
    printf("Enter interleaved string to check: ");
    scanf("%199s", a3);

    n1 = (int)strlen(a1);
    n2 = (int)strlen(a2);
    n3 = (int)strlen(a3);

    if (n3 != n1 + n2) {
        printf("The string is NOT formed by interleaving the given strings.");
        return 0;
    }

    int dp[201][201] = {0};
    dp[0][0] = 1;

    for (int i = 0; i <= n1; i++) {
        for (int j = 0; j <= n2; j++) {
            if (i == 0 && j == 0) continue;

            int k = i + j - 1;

            if (i > 0 && dp[i-1][j] && a1[i-1] == a3[k]) {
                dp[i][j] = 1;
            }
            if (j > 0 && dp[i][j-1] && a2[j-1] == a3[k]) {
                dp[i][j] = 1;
            }
        }
    }

    if (dp[n1][n2]) {
        printf("The string is formed by interleaving the given strings.");
    } else {
        printf("The string is NOT formed by interleaving the given strings.");
    }

    return 0;
}
```

Program Input (stdin)

Enter Input if required...

Output

```
Enter first string: Enter second string:
Enter interleaved string to check: The
string is formed by interleaving the
given strings.
```

Run

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Result Available
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QUESTIONS

LEVEL 2 — MEDIUM

Q1
10arrayQ2
structureQ3
stringQ4
2Darray

4 / 4 saved

Level 2 — Medium: triangular matrix (M) — 2Darray

Write a program to find the upper triangular matrix of a given matrix.

Sample Input:

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Upper triangular matrix:
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34     }
35
36     return 0;
37 }
```

Program Input (stdin)

Enter input if required...

Output

Enter number of rows: Enter number of
columns: Enter elements:
1 2 3
Upper triangular matrix:
1 2 3

Run

Save

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